| INCH-POUND |

MIL-W-29158B(YD)
11 July 1989
SUPERSEDING
MIL-W-29158A(YD)
25 February 1982

MILITARY SPECIFICATION

WINCHES, DRUMS, POWER-OPERATED, DIESEL-ENGINE-DRIVEN

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers integrally constructed diesel-engine-driven winches with pneumatic controls for general marine, and shoreline service.

1.2 Classification. The winches covered by this specification will be of the following types, as specified herein (see 6.2):

Type I - 2 drums.
Type II - 3 drums.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3950

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SPECIFICATIONS

FEDERAL

RR-W-410 - Wire Rope and Strand.
TT-P-664 - Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
UU-T-81 - Tag, Shipping and Stock.
VV-F-800 - Fuel Oil, Diesel.
VV-L-751 - Lubricating Oil, Chain, Wire Rope, and Exposed Gear.
PPP-P-40 - Packaging and Packing of Hand Tools.
PPP-T-60 - Tape, Packaging, Waterproof.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-L-2105 - Lubricating Oil, Gear, Multipurpose.
MIL-C-3600 - Compressors, Air and Gas (Except Oxygen), Packaging of.
MIL-T-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-E-10062 - Engine, Preparation for Shipment and Storage of.
MIL-T-22085 - Tape, Pressure Sensitive, Adhesive, Preservation and Sealing.
MIL-B-22191 - Barrier Material, Transparent, Flexible, Heat Sealable.

STANDARDS

MILITARY

MIL-STD-209 - Slinging and liedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-2073-1 - DOD Materiel Procedures for Development and Application of Packaging Requirements

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

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2.1.2 Government drawing. The following Government drawing is a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

NAVAL FACILITIES ENGINEERING COMMAND (NAVFAC)

816449 - N.L. Equipment Winch Support, Mark WT2 and Mark WT3.

(Copies of drawing required by contractors in connection with specific acquisition functions, should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which is current on the date of the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Commercial Packaging, Practice for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The winch shall be complete with built on control station, air powered controls for brakes, clutches, diesel engine and controls, drum rotation power train with reversing manual transmission, engine powered air compressor, and air power control system. When specified (see 6.2), the winches shall be mounted on a skid base. When specified (see 6.2), winches shall be mounted on a support frame conforming to NAVFAC Drawing No. 816449, Mark W12 for type I and Mark WT3 for type II.

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3.1.1 Type I. Type I winch shall have two drums arranged in stepped (waterfall) formation. The control station shall face the winch while providing a clear view of cable operations for both the upper and lower drum rotation and of the extended cable operations, both to the right and left of the operator. The right hand drum shall have overwind wire rope payout. The left hand drum shall underwind wire rope payout threaded under the engine and air compressor assembly, and over the winch support frame cross members for deck level in hauling. A gypsy head shall be provided on the end of the right hand drum shaft on the opposite side of the winch from the control station.

3.1.2 Type II. Type II winch shall have three drums in stepped formation. The diesel engine and air compressor assembly shall be mounted at the end of the frame behind the high drum. The control station shall be mounted with the drums to the left of the operator, and shall face in the direction of the low drum. The upper and intermediate winch drums shall have overwind wire rope payout toward the front (low drum end) of the winch. The lower winch drum shall have underwind wire rope payout threaded below the drums and the engine and air compressor assembly, and above the winch frame cross members, toward the rear (engine end) of the winch for deck level inhauling. One gypsy head shall be provided on each of the lower intermediate drum shaft ends on the opposite side of the winch from the control station.

3.2 First article. When specified (see 6.2), the contractor shall furnish one complete winch of each type specified for first article inspection and approval (see 4.3 and 6.3).

3.3 Standard commercial product. The equipment shall, as a minimum be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the unit being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

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3.4.1 Wire rope. Wire rope shall be uncoated 6 x 19 strand (type I, class 2) or 6 x 37 strand (type I, class 3), Warrington-Seele Construction, improved or extra improved plow steel with independent wire rope center, conforming to RR-W-410. Unless otherwise specified (see 6.2), type I winch shall be furnished with no less than 1,000 feet of 7/8-inch nominal diameter wire rope on each drum; type II winch with no less than 2,000 feet of rope on each drum, with 1-1/8-inch diameter on the lower and intermediate drum and 7/8-inch diameter on the high or upper drum.

3.5 Performance. Operation of air controls shall have a smooth positive response, and shall give a feel of precise control to the operator when braking and clutching. The drum locking ratchet or dog controls shall seat firmly, in the released and in the set positions, and shall not be subject to accidental disengagement. The engine shall run steadily at any throttle control setting, and speed change shall be easily coordinated when clutching against load, and during dynamic load lowering operations. Failure of air supply shall not result in accidental dropping of the load. Each drum for type I winch shall be capable of maintaining a single drum line pull of not less than 30,000 pounds (lb), at a line speed of not less than 35 feet per minute (fpm), type II winch shall be capable of maintaining a single drum line pull on either drum of not less than 33,000 lb, at a line speed not less than 150 fpm. Transfer of load between the drums, with momentary load on all drums, shall be performed smoothly without noticeable reactionary shock, by use of clutch, brake, and drum holding ratchet and pawl or drum holding dog action. With not less than 2,000 feet of wire rope on the high drum, the type II winch shall be capable of maintaining a single drum line pull of not less than 15,000 lb, at a line speed not less than 300 fpm.

3.6 Design and construction. The winch and all accessories shall be integrally assembled for maximum compactness and minimum weight. The design and construction of the winch shall be in accordance with the best engineering practice. The equipment design and accessory installations shall provide accessibility for maintenance and service in the field. The design shall be such as to prevent conditions which may be hazardous to personnel or deleterious to equipment. Maximum use of antifriction roller and ball bearings shall be made. The winches shall be designed for continuous all weather use, and shall not be damaged by occasional salt water splash.

3.6.1 Control station. A control station with banked lever air powered controls shall be provided. Hand operated levers shall be provided to control engine speed, clutches, and drum brakes. The hand operated throttle control shall remain in the position set until readjusted. Clutch controls shall be provided with detents in the fully released and fully set positions. The control station shall be so situated that the operator shall have an unobstructed view of both drums and cable operation.

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3.6.2 Drums and gypsy heads. Drums and gypsy heads shall be of cast semisteel, cast steel, or alloy steel construction. Drums shall have heavily reinforced barrels and flanges, heat treated and machined, for balance and even spooling. Drum flanges shall incorporate finished brake and clutch friction surfaces. Cooling fins for maintaining balanced operating temperature shall be incorporated. The drum flange diameter and contour shall be adequate to provide operation clearance and storage capacity for not less than the specified nominal diameter and length of wire rope, or of the manufacturer's standard length, whichever is longer.

3.6.3 Winch drum clutches. Winch drum clutches shall be of the double faced friction band, diaphragm actuated multiple disk type, cone type, or of the single or external contracting type with two bands for balanced forces and short wrap to minimize self-energization. Composition friction block wearing surfaces and simple means for adjustment shall be provided. The clutches shall have air powered engagement, and spring release. Operation under maximum loading conditions for a continuous period of not less than 30 minutes shall not cause burning,nor damage to the lining when operated in accordance with the manufacturers standard procedure.

3.6.4 Winch drum brakes. Winch drum brakes shall be either the cone or the flat band type, and shall operate against a steel or cast iron finished brake ring fitted to the drum flange. Brakes shall be spring set with air powered release, except that when an automatic safety brake provision is incorporated to prevent backward drum rotation on air supply failure, at the manufacturer's option, the brakes may be air power set and spring released. Brakes shall not slip when required single drum line pull is applied tangent to the pitch circle of the first layer of wire rope on the drum. There shall be no sign of eccentricity, nor change in clearance between brake band and brake ring when the drum is rotated freely, after 30 minutes of operation of the winch under full load conditions.

3.6.5 Drum holding ratchets and pawls or dogs. Ratchets or dog engaging wheels shall be cast integral with the drum, and shall have teeth or sockets of uniform contour, to provide maximum surface contact with the pawl or dog for the transference of shock loading with a minimum of strain. The pawls or dogs shall be spring released with air powered engagement or air released with spring engagement. Control shall provide for full engagement or full release, and shall have a snap action to prevent tooth skipping and accidental corner snagging. Pawls shall show no signs of distortion or breakage when locked in position, and loaded by the application of a line pull of not less than 30,000 lb, with a rope lead tangent to the bare drum wrap level.

3.6.6 Gears, pinions, sprockets, and shafts. Gears, pinions, and sprockets, shall be of forged or heat treated steel, or cast alloy steel, of a structurally adequate alloy suitable for waterfront use. Rotating shafts shall be machined from alloy steel. All steel shall be suitable for seaside application.

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3.6.7 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.7 Diesel engine. The diesel engine shall have horsepower, torque, and speed characteristics to meet satisfactorily all performance requirements specified herein. The engine shall start within 5 minutes and be ready for full load operation within 15 minutes in ambient temperature from +125° Fahrenheit (F) to -20°F. Engine shall be furnished complete with the following:

a. A 12-volt electrical cranking system. Battery shall be 12-volt potential, negative ground. The voltage and ampere-hour rating of the batteries shall be sufficient to provide not less than 5 minutes of total intermittent engine cranking time while retaining a battery voltage of not less than 1.0 volts per cell. Battery shall be shipped charged and dry.
b. A battery charging alternator, with a rating of not less than 35 amperes.
c. An intake air cleaner.
d. Hour meter.
e. Lubricating oil pressure indicator.
f. Cooling media temperature indicator.
g. Start-stop control.
h. Battery-alternator indicator.
i. Muffler.
j. Fuel tank level indicator.
k. Fuel tank with sufficient capacity for not less than 8 hours of normal operation.
l. Engine housing of weather resistant heavy-gauge sheet steel. The housing shall be removable or otherwise designed to permit engine removal and replacement.

3.7.1 Diesel engine fuel. When specified (see 6.2), the commercial diesel engine supplied shall provide the power necessary to meet the total performance requirements specified herein using fuel conforming to MIL-T-5624, grade JP-5 as well as fuel conforming to VV-F-800, grade DF-2.

3.8 Power transmission. The transmission shall be manual type with not less than 3 speeds forward, 1 speed reverse, and a neutral position for free payout of line.

3.9 Air compressor. Air compressor shall be the type with sufficient capacity to operate the winch air power control system. Air compressor shall include a receiver, controls, and gages.

3.10 Base, skid-type. The winch shall be mounted on the manufacturer's designed skid base consisting of not less than two full length steel channels.

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3.11 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.12 Lifting and tiedown attachments. The unit shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform type III of MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically attached to the unit. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments may be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tiedown of the unit on the carrier when shipped.

3.13 Toolbox. When specified (see 6.2), a toolbox shall be provided. The toolbox shall be large enough to store all tools required for field service or maintenance, but shall not have external closed dimensions less than 14 inches in length, 6 inches in width, and 6 inches in height. The toolbox shall be of a nominal 0.0747-inch (U.S. revised standard gage No. 14) thick steel with a hinged lid and a trunk drawbolt to keep the lid secure when vibrated. The toolbox shall be mounted in a protected, accessible location.

3.14 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as soil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

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3.15 Workmanship.

3.15.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.15.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lock washers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.15.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.15.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.15.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

3.16 Instruction plates. The equipment shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.17 Identification plate. An identification plate will be furnished by the contracting officer for each winch. The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose, and securely affix it to each unit in a conspicuous place with non-ferrous screws, rivets, or bolts not less than 1/8-inch in diameter. The applicable nomenclature contained in contract item description shall be placed in the top blank.

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3.18 Servicing and restoration. Each unit tested shall be serviced and restored to a service condition equal to the original condition of the unit, neglecting nominal wear incurred during the tests. The restoration shall include paint touchup or repainting, as required for delivery.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. The first article inspection shall be performed on one unit when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.5 and the tests of 4.6. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the winch meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining winches to be furnished under the contract.

4.4 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.5, the test of 4.6.1, and the packaging inspection of 4.7. This inspection shall be performed on each production unit.

4.5 Examination. Each unit shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.6 Tests. The first article shall be tested as specified in 4.6.1, 4.6.2, and 4.6.3. Each production unit shall be tested as specified in 4.6.1. Failure to pass any test shall constitute cause for rejection.

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4.6.1 Operational test. The operational tests shall include, but not limited, to those specified herein. Failure of any component to meet requirements, as demonstrated by the appropriate test, shall constitute failure of this test.

a. Advance engine speed to the manufacturer's recommended continuous speed.
b. Engage drum clutch to determine smoothness of operation and tendency to grab.
c. Release clutch and allow drum to spin freely to determine balance, end play, alinement, and tendency to bind or chatter.
d. Re-engage the clutch, then release, and brake drum to a stop to determine smoothness of brake action.
e. Reduce engine speed to idle.
f. Hand turn the drum and engage the pawl to determine fit with the ratchet and any binding in the control mechanism.
g. Rotate drum by hand sufficiently in both directions, to determine binding, drag from the rotating shaft, free clearance of clutch and brake, and fit between the pawl and each tooth of the ratchet.
h. Check for signs of overheating and irregular governor action.
i. Advance engine speed to maximum for approximately 15 seconds, and retard to continuous speed.
j. Repeat the series not less than five times or continue test for 25 minutes.

4.6.2 Lift test. This test shall be conducted by placing lifting cables to the lifting attachments (see 3.12). Lift the unit with a crane and hold for at least five minutes. Evidence of permanent deformation or distortion of any structural member or damage to any component shall constitute failure of this test.

4.6.3 Load tests. This test shall be conducted using actual or simulated loading, to verify compliance with the requirements of this specification.

a. Bolt the winch to a suitable foundation, and install the applicable length and size wire rope on the drums.
b. Apply a static line pull of 33,000 lbs to a wire rope lead tangent to the periphery of a full drum barrel first layer of rope, and observe for signs of brake slipping, pawl and drum misalignment, and any other distortion, under the following conditions:

    1. Drum pawls or holding dogs engaged and brakes released.
    2. Drum brakes set and pawls or holding dogs released.

c. Type I winch - With the required size cable on the first wrap of both drums, and drums turning at the equivalent of 35 fpm line speed, measured line pull to be not less than 30,000 pounds.
d. Type II winch - With the required size cable on either the intermediate and lower drum, a rope lead tangent to a periphery of between the ninth and eleventh layer on the drum, at any line speed between 150 and 200 fpm, measure a line pull not less than 33,000 lb. With the required size rope on the upper drum, with a lead tangent to a periphery between the eighth and ninth layer, and a line speed between 300 and 400 fpm, measure a line pull not less than 15,000 lb.

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4.7 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly and matchmarking. Disassembly shall be the minimum necessary to safeguard parts known to be subject to damage or loss and to accomplish reduction in cube. Bolts, nuts, screws, pins, and washers removed shall be reinstalled to one of the mating parts and secured to prevent their loss. When necessary to facilitate reassembly, parts removed and mating parts shall be matchmarked. Matchmarking shall be printed on waterproof tags, complying with UU-T-81, attached to mating parts. Waterproof ink shall be used for marking of information on tags.

5.1.1.2 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation specified in the following paragraphs are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

5.1.1.3 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

5.1.1.4 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 and not specifically provided for herein shall be preserved as follows:

5.1.1.4.1 Unfinished (not machined) surfaces. Unfinished exterior metal surfaces shall be coated with type P-1 preservative.

5.1.1.4.2 Machined surfaces. Exposed machined surfaces shall be coated with type P-6 or P-11 preservative and wrapped or covered, as applicable, with barrier material conforming to MIL-B-121, type I, grade A, class 2 or MIL-B-22191, type II. The material shall be secured in place with waterproof tape conforming to PPP-T-60 or MIL-T-22085.

5.1.1.5 Engines. Engines, engine components, and accessories shall be preserved in accordance with the level A requirements of MIL-E-10062, type I, method 1 for uncrated equipment and type II, method 1 for crated equipment.

5.1.1.6 Transmissions and torque converters. The housings shall be filled to the operating level with the lubricant required for operation. The transmissions and torque converters shall be operated to insure coating of all interior surfaces and parts with the lubricant.

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5.1.1.7 Clutches.

5.1.1.7.1 Disk type. With the cover plates removed and the clutch engaged, all accessible metal interior components of the clutch shall be sprayed with a thin film of primer conforming to TT-P-664. Coating of nonmetallic surfaces shall be kept to a minimum. Clutch control mechanisms not enclosed shall be coated with type P-1 preservative. The clutch shall be secured in the completely disengaged position.

5.1.1.7.3 Cone type. The unlined cone and clutch control mechanisms shall be coated with a thin film of primer conforming to TT-P-664. The clutches shall not be operated after application of the primer. Clutch control mechanisms not enclosed with housings shall be coated with type P-1 preservative. The clutch shall be released to prevent the lining from adhering to the unlined cones.

5.1.1.7.3 Band type. Clutch drum facings shall be coated with a thin film of primer conforming to TT-P-664. Clutch control mechanisms enclosed in the housing shall be fogged with type P-10, grade 10, preservative. Preservatives shall not contact the clutch lining.

5.1.1.8 Gears.

5.1.1.8.1 Exposed gears. Exposed gears shall be coated with type P-1 preservative or a thin film of primer conforming to TT-P-664.

5.1.1.8.2 Enclosed gears. Gear housings containing lubricant conforming to the applicable lubrication order, or to MIL-L-2105, shall be checked for proper operating level and the gears actuated to insure coating of all interior parts and surfaces. Gear housings not containing these lubricants shall be drained and refilled to the operating level with lubricant conforming to the applicable lubrication order and actuated as stated above.

5.1.1.9 Drive chains.

5.1.1.9.1 Enclosed chains. Enclosed chains and chain housings shall be preserved as specified for enclosed gears.

5.1.1.9.2 Exposed chains. Exposed chains shall be coated with enough type P-3 or P-9 preservative to assure penetration of the preservatives to the inner surfaces of the rollers, pins, and bushings. After the excess preservative has drained, the chains and sprockets shall be coated with type P-1 preservative.

5.1.1.10 Brakes. Brake drum facings shall be coated with a thin film of primer conforming to TT-P-664. Care shall be exercised to prevent the primer from coming in contact with the composition lining. Unpainted exposed surfaces of control mechanisms shall be coated with type P-1 preservative.

5.1.1.11 Hydraulic system. The hydraulic fluid supply tanks shall be filled to the operating level with hydraulic fluid required for operation.

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5.1.1.12 Air compressor. The air compressor and air compressor accessories shall be preserved in accordance with level A requirements of MIL-C-3600.

5.1.1.13 Wire rope. Unreel and apply lubricant conforming to VV-L-751, grade 2 to all surfaces so that the rope is completely immersed.

5.1.1.14 Lubrication. Lubricate all grease fittings with type P-11 preservative. Lubricate control mechanisms, linkage, hinges, fasteners, and all pivot points with P-9.

5.1.1.15 Maintenance tools. Maintenance tools shall be preserved and packaged in accordance with level A of PPP-P-40.

5.1.1.16 Technical publications. Technical publications for each winch shall be preserved method IC-1 or IC-3, in accordance with MIL-P-116.

5.1.1.17 Consolidation. The tool box shall be utilized for the consolidation of loose components, tools, and publications. The tool box lid shall be closed and secured. The remainder of components that will not fit in the tool box shall be consolidated in boxes conforming to the appropriate level A box listed in appendix C, table VII of MIL-STD-2073-1. Strapping, or reinforcement, of boxes shall not be required of containers which will be placed in closed crates.

5.1.2 Level B. Level B preservation requirements shall be the same as level A with the following exceptions:

5.1.2.1 Engines. The engine shall be preserved as specified for level A except that the fuel system, combustion chambers and valves, governor, and generator and starter shall not require preservation.

5.1.2.2 Clutches. The clutch shall not require preservation.

5.1.2.3 Brakes. Brakes shall not require preservation.

5.1.2.4 Consolidation. Requirements shall be the same as for level A except that for uncrated winches, the appropriate level B box listed in appendix C, table VII of MIL-STD-2073-1.

5.1.3 Commercial. Winches shall be preserved in accordance with the requirements of ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Crated winches. Winches weighing less than 16,000 lb shall be packed in a crate conforming to appendix C, table VII, MIL-STD-2073-1 for a wood crate, lumber and plywood sheathed, nailed and bolted, with a max weight of 30,000 lbs. Consolidation containers shall be blocked and braced within the crate.

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5.2.1.2 Uncrated winches. Winches weighing 16,000 lb and over shall be shipped uncrated in accordance with 5.2.3. Consolidation containers shall be secured to the equipment so as not to interfere with lifting.

5.2.2 Level B.

5.2.2.1 Crated winches. Winches weighing less than 16,000 lb shall be packed in a crate conforming to appendix C, table VII, MIL-STD-2073-1, for an open wood crate, bolted or nailed assembly, with a max weight of 16,000 lbs. Consolidation containers shall be blocked and braced within the crate.

5.2.2.2 Uncrated winches. Winches weighing 16,000 lbs and over shall be packed as specified for level A.

5.2.3 Commercial. Each complete winch shall be packed in accordance with ASTM D 3951.

5.3 Marking. Marking shall be in accordance with the requirements of ASTM D 3951.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The winches covered by this specification are used in pontoon barge assemblies, warping tugs, gate vessels, derricks for anchor service, pile-driving operations, bottom laid fuel pipeline launching operations, and in various shore applications.

6.2. Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Type of winch required (see 1.2).
d. When skid base is required (see 3.1).
e. When NAVFAC support frame is required (see 3.1).
f. When first article is required for inspection and approval (see 3.2 and 4.3).
g. When size and minimum length of wire rope is other than specified (see 3.4.1).
h. When engine functioning on JP-5 and DF-2 is required (see 3.7.1).
i. When fungus resistance is required (see 3.11).
j. When toolbox is required (see 3.13).
k. Color of finish coat required (see 3.14).
l. Level of preservation and level of packing required (see 5.1 and 5.2).

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6.3 First article. When a first article inspection is required, the item shall be tested and should be a first production item consisting of one complete unit, or it may be a standard production item from the contractor's current inventory as specified in 4.3. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data should be delivered by the contractor in accordance with the contract or purchase order requirements.

6.5 Subject term (key word) listing.

Compressor, air
Diesel engine
Marine Service
Skid base
Three drums
Two drums
Wire rope

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Navy - YD

Preparing activity:
Navy - YD

(Project 3950-N017)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-W-29158B(YD)	MILITARY SPECIFICATION FOR WINCHES, DRUMS, POWER-OPERATED, DIESEL-ENGINE-DRIVEN

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION (Mark one)
	☐ VENDOR
b. ADDRESS (Street, City, State, ZIP Code)	☐ USER
	☐ MANUFACTURER
	☐ OTHER (Specify):

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional	b. WORK TELEPHONE NUMBER (Include Area Code) – Optional
c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional	8. DATE OF SUBMISSION (YYMMDD)

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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